Semantic Prioritization of Domain Data for Proactive Identification of Cybersecurity Indicators

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Topic & Literature Recap

- Early ML in the Cyber space required perfectly cleaned data
- Later developments improved affairs, but still needed carefully tuned data
- Fortunately, modern semantic ML is breaking new ground for indicator analysis
- As such, our team will mine semantic relevance from raw data to enrich previous ML developments with proactive data processing
- Such enrichments could prevent novel threats & improve human readability

Research Questions:

- Is there a correlation between domain's lexical and semantic metadata?
- **Do domain semantics help predict real-world malicious activity to a statistically significant degree?**

Dataset 1: Tranco

List ID: 6GPNX | Date accessed: 03/05/2026 [11]

- CSV list of the top 1 million active domains
- Designed to be "Research-Oriented"
- Only 2 included features: popularity rank and domain name
- Features extracted during data cleanup include top-level domain, domain_length, dot_count, hyphen_count, digit_count, and digit_ratio

Dataset 2: PhishTank

- Created by Cisco Talos Threat Intelligence [10], but has Community-provided submissions
- Structured list of both alleged and confirmed phishing URLs
- The same root domain can appear throughout the list as multiple URLs (i.e. `bad.com` VS `bad.com/trick` VS `bad.com/scam`)
- Many features and tens of thousands of samples, key features include: impersonation target, if the phish is verified, if the URL is online, and time series data for when the URL is submitted or verified
- Features extracted during data cleanup include `top-level domain`, `domain_length`, `dot_count`, `hyphen_count`, `digit_count`, and `digit_ratio`

Data Preparation & Lessons Learned

- Our results were multiple extracted features, 23 GB of embeddings data, and 40k+ rows of cosine similarity
- Engineering our own features was necessary, as like Figure 1 shows, the built in ones are incomplete without human review
- We cleaned unneeded info like trailing slashes and the http scheme from PhishTank URLs, which provides little semantic value
- From there, we did gemini-embedding-001 to extract semantic features. Embedding gives us a vector space understanding of our domain indicators. [12]
- We then finished off with Cosine Similarity of the vectors to condense their value into a single similarity metric. Each phish had its top 5 semantically similar Tranco domains listed.
- A huge issue that arose in ML was working around cosine similarity being just a single feature for a training set. We address that in the next slide.

Figure 13a: PhishTank - Feature Correlation

	full_length	host_length	path_depth	dot_count	hyphen_count	digit_count	digit_ratio	subdomain_depth
full_length	1.00	0.05	0.23	0.20	0.22	0.96	0.11	0.11
host_length	0.05	1.00	-0.25	0.40	0.15	0.10	0.30	0.60
path_depth	0.23	-0.25	1.00	0.17	0.31	0.18	0.12	0.07
dot_count	0.20	0.40	0.17	1.00	0.23	0.25	0.25	0.64
hyphen_count	0.22	0.15	0.31	0.23	1.00	0.27	0.20	0.16
digit_count	0.96	0.10	0.18	0.25	0.27	1.00	0.32	0.12
digit_ratio	0.11	0.30	0.12	0.25	0.20	0.32	1.00	0.20
subdomain_depth	0.11	0.60	0.07	0.64	0.16	0.12	0.20	1.00

Figure 1: Top 15 Phishing Targets

Machine Learning First Steps

```
Training set: 37,372 samples
Test set:      9,344 samples
Train class 1 ratio: 25.00%
Test  class 1 ratio: 25.00%
```

```
search = RandomizedSearchCV(
    svm_pipeline, param_distributions,
    n_iter=n_iter, cv=cv, scoring='f1',
    n_jobs=-1, verbose=1,
    random_state=random_state,
    return_train_score=True
)
search.fit(X_train, y_train)
```

- Since Cosine Similarity produces only a single feature, a full ML model would not be needed to predict that, only a regression model
- As such, we use the Cosine Similarity as a classification for a new feature, `is_impersonated`, being defined as the 75th percentile of semantically similar domains
- This new feature lets us showcase a more robust model: we use the lexical features we had from earlier to make an SVM, and train the SVM to produce `is_impersonated`
- This lets us laser-focus in on the risk of impersonation, something the `target` feature of the base PhishTank data was woefully lacking in
- We use semantically segmented data to create a classification to predict, and then can use a cheaper SVM to predict if semantic analysis is warranted for an indicator. This 2 phased approach is akin to the Cyber concept of defense in depth.

Metrics Showcase

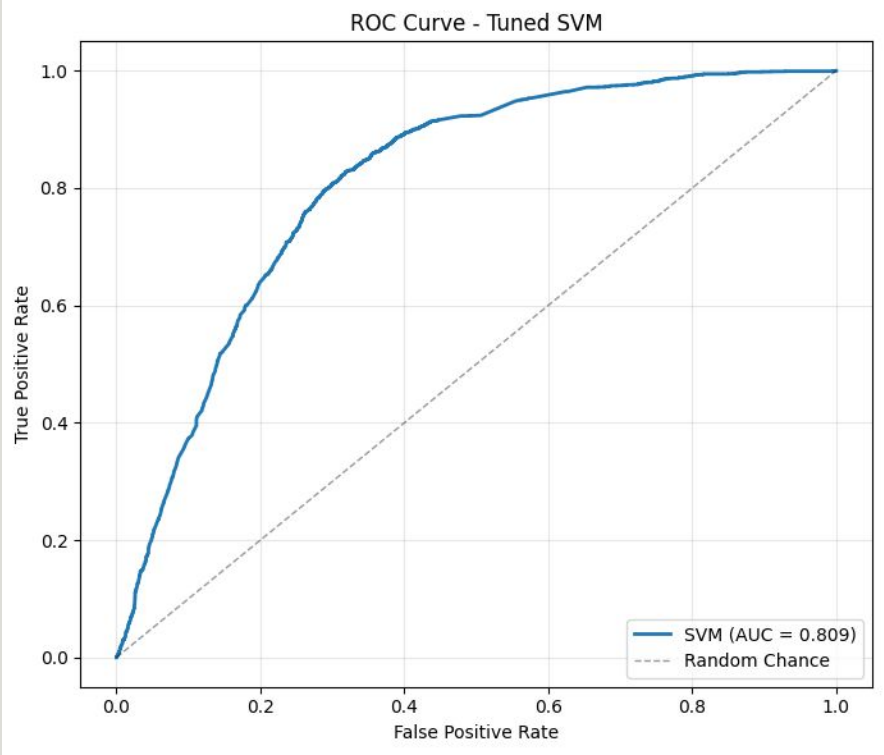
```
Tuned SVM (RBF) - Test Set Results
                   precision    recall  f1-score   support

Not Impersonating       0.93      0.64      0.76      7008
    Impersonating       0.44      0.86      0.58      2336

         accuracy                           0.69      9344
        macro avg       0.69      0.75      0.67      9344
     weighted avg       0.81      0.69      0.71      9344
```

```python
def train_baseline_svm(X_train, y_train, X_test, y_test, random_state):
    """Train a baseline SVM (RBF kernel) and print test-set classification report."""
    svm_pipeline = Pipeline([
        ('scaler', StandardScaler()),
        ('svm', SVC(kernel='rbf', class_weight='balanced', probability=True,
                    random_state=random_state))
    ])
    svm_pipeline.fit(X_train, y_train)
    y_pred = svm_pipeline.predict(X_test)

    print("Baseline SVM (RBF) - Test Set Results")
    print(classification_report(y_test, y_pred,
                                target_names=['Not Impersonating', 'Impersonating']))
    return svm_pipeline, y_pred
```



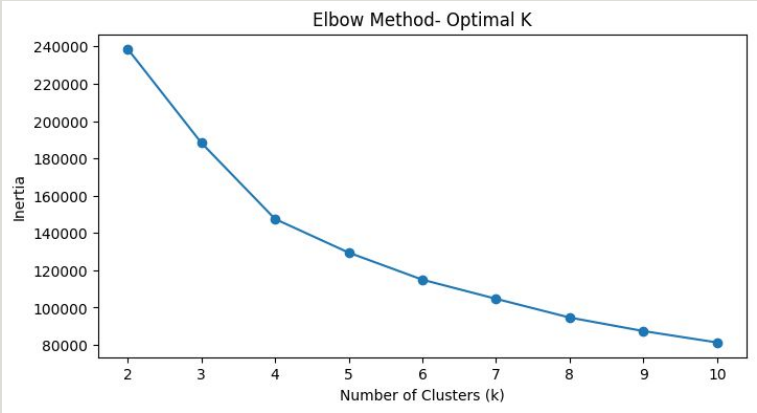
Results: Metrics Interpretation

- The ML model agrees with our intuitive thoughts on URLs: longer, more complex URLs try to exhaust the user, while shorter ones often rely on tactics like typosquatting
- We use random searching for hyperparameter tuning to automatically create the best tune for our dataset
- Our initial SVM has an AUC above 0.5, meaning we have beaten random chance and reached into true predictions
- In its totality, we were able to get a tuned SVM classifier to achieve meaningful discrimination impersonated and legitimate domains by combining lexical and semantic processing
- As a major weakness, the 75th percentile threshold is not trained or refined by any ML analysis, meaning we are relying on the accuracy of the pre-trained embedding framework on classifying similarity well

Unsupervised Clustering - Phishtank

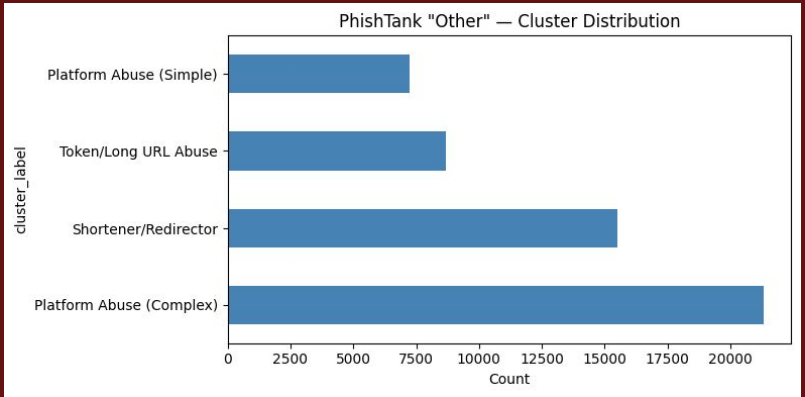
94% of Phishtank targets are labeled other; what can we identify about these?

- K-Means clustering applied to standardized lexical features to find common structures
- Elbow method applied to choose optimal number of categories to avoid introducing noise or overfitting
- Features used: host length, path depth, dot count, hyphen count, digit ratio, subdomain depth.



Results: Four Phishing Archetypes

- Platform abuse (simple and complex) is most common: Google Sites, Wix, Weebly
- Link shorteners: data agrees that it is correct to be suspicious of them!
- Google Docs, Dropbox: abuse via long token strings



Examples by Cluster

Sample URLs by Cluster

Cluster	Example URL
Platform Abuse (Simple)	https://pub-d637c8d427db484db2245748126bd788.r2.dev/rcubic.h...
Platform Abuse (Simple)	https://reserved-team-373417.framer.app/
Platform Abuse (Simple)	https://pub-cec2c769e7424225875847093b63e672.r2.dev/index.ht...
Platform Abuse (Complex)	https://please-wait--connection.firebaseapp.com/
Platform Abuse (Complex)	https://wxooontreonr.blogspot.com/
Platform Abuse (Complex)	https://startus-home.ghost.io/live-enus/
Token/Long URL Abuse	https://docs.google.com/presentation/d/e/2PACX-1vSd2kgD9d1Ub...
Token/Long URL Abuse	http://zuza.stefek.szczecin.pl/Webmail/75/Webmail/webmail.ph...
Token/Long URL Abuse	https://docs.google.com/presentation/d/e/2PACX-1vTMeGlpqWyMI...
Shortener/Redirector	https://qrco.de/bgbcV6
Shortener/Redirector	https://i8.ae/cHKFq
Shortener/Redirector	https://tinyurl.com/3xht4cub

Next Steps

- Implement research guidelines into data interpretation sections
- Strengthen verbiage in our final paper
- Work on final presentation and clean Jupyter Notebook
- Finish tweaking ML model to a level where we are happy

Works Cited

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